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Periodontal treatment modulates gene expression of endothelial progenitor cells in diabetic patients

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Running head: Periodontal therapy modulates EPC genes

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KEYWORDS

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Clinical Relevance

Scientific rational for the study: Our previous study shows that periodontal treatment decreases the counts of endothelial progenitor cells (EPCs) in systemically healthy subjects. This study extended to investigate the effects of periodontal treatment on the immuno-inflammatory gene expression of EPCs in patients with type 2 diabetes mellitus (T2DM).

Principal findings: Periodontal treatment contributed to a notable reduction of immuno-inflammatory response measured by IL-6 and IL-8 transcription in EPCs, in parallel with the improvement of periodontal conditions.

Practical implications: The preliminary study suggests that periodontal intervention may favorably affect the gene expression of EPCs isolated from peripheral blood in patients with T2DM, through tackling periodontal infection and inflammation as well as modulating systemic level of inflammation. Improvement of periodontal health could be beneficial to diabetes care.

Conflict of interest and source of funding

The authors declared that they have no conflict of interest.

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Abstract

Background: Periodontitis is closely linked with type 2 diabetes mellitus (T2DM) and endothelial dysfunction. This study investigated the effects of periodontal treatment on immuno-inflammatory gene expression of endothelial progenitor cells (EPCs) in diabetic patients.

Methods: Eighteen T2DM patients with moderate to severe chronic periodontitis were randomly assigned to the Treatment group with oral hygiene instruction (OHI), scaling and root debridement (n=11), and Control group (n=7) with OHI alone. Peripheral blood samples were taken for biochemical analysis and culture of EPCs at baseline and 6 months after the treatment. PCR array-based profiling of 84 Toll-like receptor signaling-related genes in EPCs was firstly assessed for four randomly selected patients from the Treatment group. The differentially expressed genes were then further validated by qPCR in both groups.

Results: All subjects in the Treatment group significantly improved their periodontal conditions. Among the 84 genes tested, IL-6 and IL-8 transcripts were significantly down-regulated with over 2-fold change after the treatment, and this observation was further validated by qPCR in all subjects from both groups ($p<0.05$).

Conclusion: This preliminary study suggests that periodontal treatment may contribute to a notable reduction of immuno-inflammatory gene expression measured by IL-6 and IL-8 transcripts in EPCs.

The study was approved by the Institutional Review Board of the University of Hong Kong/Hospital Authority of Hong Kong West Cluster (IRB UW 07-165) and registered at http://www.hkuctr.com (Study Identifier: HKUCTR-458).

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INTRODUCTION

Non-communicable diseases (NCDs) including periodontitis remain the major global health burden and crisis with huge socio-economic impacts (Chapple, 2014; Daar et al., 2007; Jin et al., 2016; Kassebaum et al., 2014). Diabetes mellitus (DM) is one of the highly prevalent NCDs, and it is closely linked to periodontitis via a defined two-way relationship (Grossi & Genco, 1998). Periodontitis may aggregate inflammatory conditions locally and systemically, and worsen metabolic control, contributing to development of cardiovascular disease (CVD) and DM complications (Nitta et al., 2017; Saremi et al., 2005; Southerland et al., 2012; Taylor et al., 1996). Effective control of periodontal diseases significantly account for reduction of systemic level of inflammation and improvement of glycemic control (Chen et al., 2012; Merchant et al., 2016). Furthermore, periodontitis is closely associated with endothelial dysfunction, and periodontal treatment could improve endothelial function in non-DM patients (Holtfreter et al., 2013; Seinost et al., 2005; Tonetti et al., 2007), although the underlying mechanism remains unclear.

Endothelial cells play a critical role in vascular homeostasis, not only as a barrier but also a signal transducer for circulating risk factors that modify the vascular phenotype (Woywodt et al., 2002). The endothelial integrity depends on the replication of adjacent mature endothelial cells, and appropriate function of circulating endothelial progenitor cells (EPCs) for repair (Deanfield et al., 2007). Since the first isolation of EPCs (Asahara et al., 1997), it has been well noted that the circulating EPCs originated from bone marrow enable to promote endothelial repair and maintain vascular integrity (Burger & Touyz, 2012). The number and function of EPCs may serve as useful surrogate biological markers for monitoring the status of CVD and DM (Fadini et al., 2005; Hill et al., 2003; Werner et al., 2005). EPCs could be identified by fluorescence-activated cell sorting (FACS) of blood cells with surface markers of CD34, CD133 and kinase insert domain-containing receptor (KDR) (Wojakowski et al., 2008), and cell culture of fibronectin-adherent peripheral blood mononuclear cells in endothelial cell-specific media. The latter approach yields two distinct types of EPCs depending on the time of emergence, i.e. spindle-shaped early EPCs within one week of culture, and cobblestone-like late EPCs at 2 to 3 weeks (Hur et al., 2004). Early EPCs (i.e. monocytic EPCs)

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with limited proliferation ability critically contribute to angiogenesis via a paracrine manner by secreting pro-angiogenic growth factors. Whereas, late EPCs (i.e. endothelial colony forming cells) with highly proliferative ability and committed endothelial lineage can expand into mature endothelial cells and thus integrate into the vasculature (Yoder, 2012).

We have previously found that periodontitis is associated with increased levels of $CD34^+/KDR^+$ EPCs, and periodontal treatment significantly decreases $CD34^+$ progenitor cell counts in systemically healthy subjects (Li et al., 2009; 2011). A recent study further supports that EPCs counts are related to severity of periodontitis (Jonsson et al., 2014). In these studies, the primary outcome measure is the counts of EPCs, while their genetic and biological profiles are not assessed. In addition, dysfunction of EPCs has been demonstrated to contribute to diabetic complications (Tepper et al., 2002). As Toll-like receptors (TLRs) play a critical role in recognition of pathogen-associated molecular patterns (PAMP) in host immuno-inflammatory response (Akira et al., 2006; Morris et al., 2006), and TLR-mediated inflammatory cytokines/chemokines are involved in the paracrine pattern of EPCs, this preliminary study extended to investigate the effects of periodontal treatment on TLR-related gene expression of EPCs in patients with type 2 DM (T2DM).

MATERIALS AND METHODS

Subjects

A total of 41 T2DM patients defined by WHO criteria were recruited from the Department of Medicine, Queen Mary Hospital, Li Ka Shing Faculty of Medicine, The University of Hong Kong between June 2015 and August 2016. All patients with documented history of CVD were excluded from the study. Smoking status was recorded but smokers were not excluded. The inclusion criteria included i) clinical diagnosis of DM for at least 5 years with H1bAc level more than 6.5%; ii) consistent anti-diabetic treatment 3 months prior to the study; and iii) at least 40 years old. Patients were periodontally screened in Prince Philip Dental Hospital (PPDH), and the following inclusion criteria for moderate to severe chronic periodontitis (Li et al., 2009) were met, including over 6 sites

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with probing depth (PD) ≥ 4mm and over 25% of sites with interproximal clinical attachment loss (CAL) ≥ 5mm as well as at least 10 teeth present. Patients with antibiotic/anti-inflammatory treatment within three months prior to the study or those requiring antibiotic prophylaxis were excluded. The flow diagram of the study is presented in Fig. 1.

After screening in PPDH, 18 subjects were recruited and randomized into the Treatment (n=11) and Control groups (n=7) according to the ratio of 3:2. A restricted randomized approach was conducted to prevent the imbalance of age, gender, DM duration and severity of periodontitis (Altman & Bland, 2005). The primary investigator generated the allocation sequence and concealed the sequence. One examiner performed periodontal examination and the other one carried out the medical tests, and both were blind to the group assignments. The Treatment group received non-surgical periodontal therapy immediately following the baseline examination, while the Control group got the treatment after completion of the re-assessment at 6-month follow-up. 30 mL of fasting blood were taken from each patient at baseline and 6-month. Oral and written informant consent were obtained from all participants prior to the study. The study was approved by the Institutional Review Board of the University of Hong Kong/Hospital Authority of Hong Kong West Cluster (IRB UW 07-165) and registered at http://www.hkuctr.com (HKUCTR-458).

Periodontal examination and nonsurgical periodontal therapy

Full-mouth periodontal examination was undertaken by a single calibrated examiner (H.L), having a high intra-examiner reproducibility with an intraclass correlation coefficient of 0.91 for PD and 0.89 for CAL, respectively. Clinical data including missing teeth, plaque score, bleeding on probing (BoP), PD and gingival recession (GR) were recorded at six sites per tooth excluding 3rd molars at baseline and 6-month follow-up. CAL was recorded at the appropriate sites on the basis of PD and GR (Li et al., 2009).

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Afterwards, all subjects received professional oral hygiene instruction, while non-surgical periodontal therapy was only performed for the Treatment group, including extraction of hopeless teeth, scaling and root debridement with both hand instruments and ultrasonic scaler. Review appointments were given every 4 to 6 weeks to reinforce patients' plaque control. Similar treatment was provided for the Control group after completion of the study.

Isolation and culture of EPCs

Cell isolation and culture were undertaken following an established protocol with minor modification (Medina et al., 2011). Briefly, peripheral blood mononuclear cells (PBMNCs) were isolated by gradient centrifugation (Ficoll-Paque Premium, GE Healthcare, Sweden) and plated on fibronectin coated 6-well plates (Sigma, USA) or chamber slides (ibidi, Germany) at a density of 3×10^6 cells/mL. PBMNCs were cultured in EGM-2 medium (Lonza, USA) supplemented with 10% FBS, growth factors and antibiotics. After 4 days, the non-adherent cells were gently washed away with PBS twice, and fresh medium was changed daily thereafter. Cells and culture media were harvested for further analysis on Day 7.

Characterization of EPCs

The attached cells on chamber slides were firstly assessed for the phenotype of EPCs by Dil-labeled acetylated low density lipoprotein (Dil-ac-LDL) uptake and Ulex europaeus agglutinin-1 (UEA-1) staining as previously described (Cheng et al., 2013). In brief, cells were incubated with Dil-ac-LDL (2.5ug/mL, Invitrogen) for 4 h. Afterwards, cells were fixed with 4% paraformaldehyde (PFA) and incubated with FITC–labeled UEA-1 at 10ug/mL (Vector Laboratories, USA) for 1 h. After counterstained with DAPI to show the nucleus, cells were visualized with an inverted fluorescent microscope. Dual-stained cells positive for both ac-LDL and UEA-1 were deemed as EPCs.

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FACS analysis was performed to further confirm the marker of the cultured EPCs. Cells were detached with Accutase (Gibco, USA) and incubated with the antibodies of APC-conjugated CD14, PerCP-conjugated CD45, FITC-conjugated CD31 (BD Pharmingen, USA) and PE-conjugated CD133 (Miltenyi Biotech, Germany), respectively. Isotype-matched antibodies served as negative controls. FACS was performed on Flow Cytometer (FACSVerse, BD Biosciences) with $>10^5$ events recorded. Date were presented as mean±SD from at least three independent experiments.

RNA extraction

Total RNA was extracted from EPCs and purified by RNeasy Plus mini kit (Qiagen, Germany) following the manufacturer's protocol. The concentration and purity of RNA were assessed by NanoDrop spectrometer 1000 (Thermo Scientific, USA). Total RNA with an OD 260/280 ratio of 1.9 to 2.1 was aliquoted and stored under -70□ for further use.

TLR pathway RT2 PCR profiler array

To identify the expression of candidate genes which might be affected by periodontal treatment, TLR pathway RT2 PCR profiler array was performed on four patients randomly selected from the Treatment group. Briefly, cDNA was prepared from 500 ng of RNA by using RT2 first strand kit (Qiagen, Germany) and PCR reaction was conducted on StepOne Plus RT-PCR system (Applied Bioscience, USA), according to manufacturer's instructions. Raw data were uploaded and analyzed by web-based automated software for RT2 Profiler PCR Array Data Analysis (http://www.qiagen.com/shop/genes-and-pathways/data-analysis-center-overview-page).

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RT-PCR

To validate the differentially expressed genes selected from the RT2 profiler analysis, RNAs of EPCs from all subjects were reverse transcripted to cDNAs by the QuantiTect Reverse Transcription Kit (Qiagen, Germany). The Primer Sequences were designed as follows: IL-6 F: 5'GGAGACTTGCCTGGTGAAAATC3' R: 5'GGGTCAGGGGTGGTTATTGC3'; IL-8 F: 5'GACATACTCCAAACCTTTCCACC3' R: 5'AACTTCTCCACAACCCTCTGC3'; GAPDH F: 5'TCCCTGAGCTGAACGGGAAG3' R: 5'GGAGGAGTGGGTGTCGCTGT3'. The QuantiNova SYBR Green PCR kit (Qiagen, Germany) was used following manufacture's protocol. The ΔΔCt method was applied to calculate the relative levels of gene expression (Livak & Schmittgen, 2001).

ELISA

The levels of IL-6 and IL-8 in the culture supernatant were determined by ELISA kits (R & D, USA) according to the manufacturer's instructions.

Statistical analysis

The primary outcome measure was the change of TLR-related gene expression of EPCs (e.g. pro-inflammatory cytokines), and the secondary ones included the changes of periodontal conditions and other biomarkers. The skewed gene expression fold changes were compared after log transformation. The intra-group comparison was performed by paired *t*-test. The inter-group difference was determined using chi-square, Fisher's test, independent *t*-test, Mann-Whitney U test as appropriate. Person correlation analysis was adopted to identify potential variables related to the changes in gene expression of EPCs after the treatment. Two-tailed p-value < 0.05 was considered statistically significant assessed by SPSS software (SPSS 14.0 SPSS Inc., Chicago, IL, USA).

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RESULTS

The demographic and diabetes-related characters

Of the 18 participants, one (P31) from the Control Group was lost to follow up, as the patient declined the follow-up call. During the 6-month study period, no one reported medication changes. Table 1 shows the demographic and diabetes-related characteristics at baseline. The mean age and diabetes duration of the Treatment Group were 65±8 years and 19±6 years respectively, while those of the controls were 68±3 years and 18±8 years. Seven patients from the Treatment group and 4 from the Control group exhibited at least one DM-related micro-angiopathy, with retinopathy being the most. Overall, there is no significant inter-group difference in demographic and DM-related variables at baseline.

Effects of periodontal treatment on gene expression of EPCs

All periodontal parameters were significantly improved after the treatment as compared with the Controls, e.g. sites% with BoP reduced by approximately 38%, PD ≥ 4mm decreased (26.9% to 6.0%) and CAL ≥ 5mm reduced (41.8% to 34.8%) ($p < 0.05$) (Table 2). While, no significant changes were observed in high-sensitivity C-reactive protein (hs-CRP) and hemoglobin A1c (HbA1c).

Seven days of culture in endothelial-specific media resulted in a mixed cell population with spindle-shaped morphology (Fig. 2a). Over 95% of them were double positive for ac-LDL and UEA-1 (Fig. 2b). The cell lineage markers (Fig. 2c) indicated that majority of them expressed the monocytic (CD14, 89.9±10.0%) and leukocyte markers (CD45, 94±3.0%); whereas the endothelial (CD31, 21.3±12.7%) and the progenitor markers (CD133, 18.2 ± 6.9%) were less expressed to a limited extent. These results were in accordance with previous studies that the EPCs isolated in this study were monocytic EPCs (Chang et al., 2014; Cheng et al., 2013).

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The RT2 PCR array results of 84 genes involved in the TLR pathway were presented in Fig. 3. As shown in the volcano plot (Fig. 3a), the horizontal x-axis exhibits the effect of periodontal treatment by displaying the fold change in log2 scale, while the vertical y-axis presents the statistical significance with the *p*-value in –log10 scale. Nine differentially expressed genes with fold change >2 were further listed in Fig. 3b. Of them, two downstream inflammatory genes were significantly modulated ($p< 0.05$), including IL-6 (fold change -4.12, p=0.040) and IL-8 (fold change -2.52, p=0.026). The genes expression of another two inflammatory mediators also decreased with a non-significant trend, including IL-1β (fold change -4.02, p=0.059) and CCL2 (fold change -4.43, p=0.086). As such, the mRNA expression of IL-6 and IL-8 was further validated in all subjects from both groups using qPCR. Overall, periodontal treatment significantly reduced the gene expression of IL-6 (treatment effect in log2 scale, fold change = -1.88, p=0.004), and IL-8 (treatment effect in log2 scale, fold change =-1.51, p=0.023) (Table 3). Regarding the individual profile of gene expression, both IL-6 and IL-8 genes remained unchanged in the Control group. Whereas, in the Treatment group IL-6 expression was reduced in 7 out of 11 (63.6%) subjects, and IL-8 was downregulated in 4 (36.4%) patients with over 2 folds. It was worthy to note that although the subject (P11) with both retinopathy and nephropathy complications and the single current smoker (P1) exhibited favorable periodontal treatment responses, the opposed effects on gene expression were found. The marked down-regulation of IL-6 (10.7-fold) and IL-8 expression (4.6-fold) occurred in the former, while increased gene expression was observed in the latter (Fig. 4a). Notably, the down-regulation of IL-6 gene expression at 6-month was significantly correlated with the improvement of periodontal conditions measured by BoP (r=0.556, p=0.021), PD ≥ 4mm (r=0.522, p=0.032) and CAL ≥ 5mm (r=0.677, p=0.003). Moreover, the reduced expression of IL-8 transcript was strongly correlated with gain of clinical attachment (r=0.657, p=0.004) (Fig. 4b). Whereas no significant correlation was found between alternation of gene expression and HbA1c level. Furthermore, IL-6 level in the culture supernatant was below the detection limit in all samples (9.4 pg/mL), while the change in IL-8 remained insignificant after the treatment.

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DISCUSSION

It is well recognized that reduced counts of EPCs and their functional capacities are strong indicators of endothelial dysfunction in patients with CVD or DM (Fadini et al., 2005; Schmidt et al., 2015). Systemic conditions may have varying effects on circulating EPC levels. For instance, lower EPC counts were found in rheumatoid arthritis patients (Grisar et al., 2005) and healthy subjects injected with low dose of lipopolysaccharides (Mayr et al., 2007). On the contrary, acute lung injury (Burnham et al., 2005) and bacterial pneumonia (Yamada et al., 2005) were significantly associated with an increased level of EPCs. Notably, our previous study has shown that periodontitis patients exhibit significantly higher EPC counts (Li et al., 2009). It may be premature to assume that the increased counts of EPCs in periodontitis patients account for the impaired endothelial function, as both number and functional capacity of EPCs serve as important surrogate markers of endothelial function. Our present study reveals that effective non-surgical periodontal treatment may contribute to notable reduction of immuno-inflammatory gene expression measured by IL-6 and IL-8 transcripts in EPCs from T2DM patients.

To investigate the EPC gene expression profiles, cell-culture method was adopted to isolate and expand EPCs from peripheral blood. Late EPCs are considered as the subtype for investigation of vascular remodeling, due to their high proliferative ability and committed endothelial lineage; whereas early EPCs consist of heterogenetic cell populations which augment neovascularization by paracrine manner (Asahara et al., 2011; Medina et al., 2012). In the present study, early EPCs were employed to assess the endothelial characteristics. In addition, hematopoietic properties were also retained in these cells with high expression of CD14 and CD45. Whereas, the fraction of late EPCs is extremely low, approximately 0.0001% of total MNCs in peripheral blood (Ingram et al., 2005). Usually, 50 to 100 ml of peripheral blood or a minimum of 20 ml of umbilical blood in healthy subjects are required to yield late EPCs (Ingram et al., 2004). A recent study shows that late EPCs could be obtained in only 9 out of 70 patients with acute coronary syndrome (Campioni et al., 2013). As such, the present study assessed the early EPCs for both ethical and technical concerns. It is noted that EPCs could reside in the vascular endothelia (Ingram et al., 2005), and late EPCs have recently

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been isolated from human and mouse pulmonary microvascular tissues (Alphonse et al., 2015). Our further study could therefore extend to investigate the profiles and activities of EPCs in periodontal tissues.

TLRs play critical roles in recognition of PAMP in immuno-inflammatory responses (Akira & Takeda, 2004; Morris et al., 2006). Notably, TLR4 activation by lipopolysaccharides induces inflammatory response and enhances stem phenotype of late EPCs (He et al., 2010). It is evident that pro-inflammatory cytokines like IL-6 and IL-8 act as important regulators in the orchestration of vascular function and pathological process of vascular diseases (Kofler et al., 2005). When homing to the damaged vessels, EPCs-derived pro-inflammatory cytokines or chemokines may activate gene expression in adjacent endothelial and smooth muscle cells, which in turn facilitates leukocyte recruitment and initiation of coagulation (Hanada & Yoshimura, 2002). Elevated expression of IL-6 and IL-8 genes and proteins has been well documented in atherosclerotic plaques, indicating that they may function as angiogenic mediators in plaque formation and contribute to the initiation and progression of atherosclerosis (Rus et al., 1996). A recent study provides direct evidence on IL-8 secretion by EPCs that could modulate angiogenesis in mature endothelial cells (Medina et al., 2011). In the present study, the TLR pathway-related PCR array data show that the IL-6 and IL-8 transcripts of EPCs could be markedly downregulated following effective periodontal treatment, and these effects are further validated by qPCR in both groups (Table 3). Whereas, such therapeutic effects on IL-6 and IL-8 gene expression varied among the subjects (Fig. 4), likely due to the difference in risk background and host susceptibility such as status of DM control, microvascular complications and tobacco smoking. In addition, the change of IL-6 and IL-8 levels in culture media of EPCs is not along with the downregulation of their transcripts after the treatment. In general, gene expression is often in parallel with its protein level, while such linkage may not necessarily exist due to various factors involved in transcription and protein synthesis (Pascal et al., 2008). In the present experiments, the ELISA assay results could be to different extent affected by the number of EPCs attached to culture plates under various circumstances. It has been reported that factors like physical exercise and

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anti-hyperglycemic treatment could significantly affect the number of cultured EPCs (Chang et al., 2016; Wang et al., 2006).

In the present study, no significant change is observed between the two groups in terms of inflammatory biomarker such as hs-CRP, which is inconsistent with the previous reports (Artese et al., 2015; Corbella et al., 2013). The discrepancy could be partly explained by the relatively limited sample size and data variation in the current study. A recent study suggests that the association of periodontitis with endothelial dysfunction could be independent of systemic levels of inflammation assessed by hs-CRP (Holtfreter et al., 2013). Moreover, Tonetti and co-workers (2007) shows that endothelial function can be significantly improved in parallel with periodontal conditions after treatment, while not necessarily along with the change in inflammatory biomarkers like CRP and IL-6. Currently, the exact mechanisms for the effect of periodontal treatment on the profile and functional activity of EPCs remain unclear, comprehensive measurements of systemic levels of infection burden and inflammatory biomarkers could have been undertaken to better understand the link of periodontitis-related systemic conditions with EPCs and endothelial function.

There are some limitations of the study to address. Firstly, due to lack of historical data, the preliminary nature of the study and budget constraint, one third of the subjects in the Treatment group were randomly selected for the PCR array analysis. The candidate genes that could be potentially affected by the periodontal treatment were then identified for further validation. Interestingly, among the 84 target genes IL-6 and IL-8 transcripts were significantly downregulated after the periodontal treatment. These results were validated by the qPCR analysis in all subjects from both groups. Moreover, G*Power 3.0-based sample size calculation (Faul et al., 2007) using the array dataset (fold change in log2 scale: IL-6, -2.11±1.44; IL-8, -1.33±0.58) indicates that the sample size of the present study is adequate to detect a difference in the gene expression of IL-6 and IL-8 at a significance level of 0.05 with 80% power. Whereas, it could not entirely exclude the possibility that other TLR-related genes could also be modulated by the treatment, due to the relatively small sample size of the study. Secondly, the blood samples were taken at baseline and 6-month after the treatment, and the altered gene expression of EPCs shortly following periodontal therapy might be underestimated. Moreover,

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the TLR-related protein profile of EPCs could have been assessed as well. These critical points need to be taken into consideration in future studies. Lastly, it could be interesting to compare the potential difference in the gene expression of EPCs between DM and non-DM subjects. Periodontal treatment can indeed alter gene expression of peripheral blood monocytes in non-DM subjects in connection to the anti-inflammation effects (Papapanou et al., 2007). Whereas, EPCs from DM patients exhibit impaired biological function with reference to non-DM patients (Tepper et al., 2002). Periodontal therapy can greatly improve atherosclerotic markers in patients with both periodontitis and other co-morbidities like CVD and/or DM (Teeuw et al., 2014). These findings collectively suggest that there might be different response of EPCs following periodontal treatment between DM and non-DM patients. Further studies are highly warranted to clarify this point.

Within the limitations of the study, we hereby provide evidence that effective non-surgical periodontal treatment may contribute to notable reduction of immuno-inflammatory response measured by IL-6 and IL-8 transcripts in EPCs from T2DM subjects. These findings may enhance our understanding on the effects of periodontal disease on endothelial dysfunction and the clinical implications of periodontal care for oral and general health.

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Figure legends

Fig. 1. Flow chart of the participants in the study.

Fig. 2. Characterization of cultured EPCs. Seven-day culture of EPCs under inverted microscope (a). Dual staining by florescence microscope (b). Cultured EPCs uptake Dil–AcLDL (Red) and bind to UEA-1 (Green). Cells were counterstained with DAPI to show the nucleus (Blue). Original magnification 200×. Expression of surface markers in EPCs by flow cytometry analysis (c).

Fig. 3. The volcano plot summarizes the changes in TLR pathway profiler array data in EPCs with four randomly selected subjects in the Treatment group after the treatment (a). The horizontal x axis represents the magnitude of biological effect by displaying the fold change in log2 scale, while the vertical y axis shows the statistical significance by displaying the *p*-value in –log10 scale. Genes expression with up- or down-regulated by over 2-fold after the treatment was highlighted (b). The bold data indicate the statistical difference with $p < 0.05$.

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Fig. 4. The effect of periodontal treatment on the gene expression of IL-6 and IL-8 in all participants from the Control group (Left X axis, n=6) and the Treatment group (Right X axis, n=11) (a). The Y axis indicates the fold change (in log2 scale) of gene expression after the treatment. Periodontal treatment resulted in pronounced decrease of IL-6 expression (< -2 fold) in 7 (63.6%) patients, while decreased expression of IL-8 (< -2 fold) occurred in 4 (36.4%) patients in the Treatment group. The dash line indicates fold change = 2 or -2. Linear correlation of the fold changes in periodontal parameters (BoP, PD and CAL) with the alternation of IL-6 and IL-8 transcripts after the treatment (b).

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Table 1. Demographic and diabetes-related characters of the 18 patients with chronic periodontitis at baseline

ID	Age (Years)	Male	DM duration (Years)	Micro-angiopathy*	Smoker	Insulin[§]	OHA	BMI (kg/m^2)
Treatment group (n=11)								
P1	50	Y	12	N	Y	N	Y	28.9
P10	65	Y	22	Y	N	Y	Y	26.8
P11	65	Y	11	Y	N	N	Y	24.1
P20	75	N	12	N	N	Y	N	28.8
P26	63	N	13	Y	N	Y	Y	30.0
P28	57	Y	19	Y	N	Y	Y	25.4
P30	70	Y	24	N	N	N	Y	25.2
P32	67	N	27	Y	N	Y	Y	22.2
P36	64	N	20	Y	N	Y	Y	25.3
P37	58	N	20	N	N	N	Y	27.9
P38	80	Y	27	Y	N	N	Y	29.6
Total	65 ± 8	54.5%	19 ± 6	63.6%	9.1%	54.5%	90.9%	26.7 ± 2.5
Control group (n=7)								
P3	69	N	21	Y	N	N	Y	27.1
P9	63	Y	8	N	Y	N	Y	31.2
P19	68	Y	19	Y	N	Y	N	22.0
P27	73	N	10	N	N	N	Y	26.0
P29	69	N	29	N	N	Y	Y	29.6
P31	66	Y	25	Y	N	N	Y	28.5
P33	69	N	14	Y	N	Y	Y	28.1
Total	68 ± 3	42.9%	18 ± 8	57.1%	14.3%	42.8%	85.7%	27.5 ± 2.9
p-values[φ]	NS	NS	NS	NS	NS	NS	NS	NS

The summarized data for each group are expressed as mean±SD or percentage as appropriate.

OHA: Oral hypoglycemic agents; BMI: Body mass index.

*Micro-angiopathy refers to at least one condition including retinopathy, nephropathy and peripheral neuropathy.

[§]Daily insulin injection.

[φ]The inter-group difference was assessed by independent *t*-test or Fisher's exact test as appropriate.

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Table 2. Periodontal characteristics, high-sensitivity C-reactive protein (hs-CRP) and hemoglobin A1c (HbA1c) levels at baseline and 6-month follow-up

	Treatment group（n=11）	Control group (n=6)	Treatment effect (95% CI)	*p*-values
No. of missing teeth				
Baseline	8.5 ± 5.5	8.3 ± 7.2		0.946
6-month follow up	10.0 ± 6.5	8.3 ± 7.2		0.636
Change	1.45 ± 1.69	0 ± 0	NA	**0.034***
BoP (%)				
Baseline	54.6 ± 21.5	57.5 ± 24.9		0.803
6-month follow up	16.0 ± 9.6	66.4 ± 39.3		**0.001**
Change	-38.6 ± 21.1	8.9 ± 17.3	-47.5 (-69.1 to -25.9)	**<0.001**
Plaque index (%)				
Baseline	66.0 ± 15.7	58.7 ± 13.5		0.352
6-month follow up	17.7 ± 14.0	67.9 ± 22.4		**<0.001**
Change	-48.2 ± 14.3	9.2 ± 12.7	-57.4 (-72.3 to -42.5)	**<0.001**
PD ≥ 4mm (%)				
Baseline	26.9 ± 19.1	43.3 ± 27.4		0.166
6-month follow up	6.0 ± 4.7	45.1 ± 34.5		**0.002**
Change	-20.9 ± 15.2	1.8 ± 13.2	-22.6 (-38.4 to -6.8)	**0.008**
CAL ≥ 5mm (%)				
Baseline	41.8 ± 27.8	49.6 ± 33.1		0.610
6-month follow up	34.8 ± 28.8	57.8 ± 32.5		0.153
Change	-6.9 ± 8.8	8.2 ± 7.4	-15.1 (-24.2 to -6.1)	**0.003**
hs-CRP (mg/L)				
Baseline	2.32 ± 2.72	3.46 ± 3.54		0.615*
6-month follow up	1.95 ± 2.27	2.75 ± 3.53		0.421*
Change	-0.37 ± 3.55	-0.72 ± 2.31	NA	0.366*
HbA1c (%)				
Baseline	7.96 ± 0.72	7.95 ± 0.94		0.974
6-month follow up	8.81 ± 1.06	8.35 ± 1.37		0.443
Change	0.85 ± 0.97	0.4 ± 1.82	0.45 (-0.97 to 1.88)	0.507

The data are presented as mean±SD. Statistical difference between groups was determined with independent *t*-test. *Mann-Whitney U test was performed when the data did not fit normal distribution, and the treatment effect was not determined.

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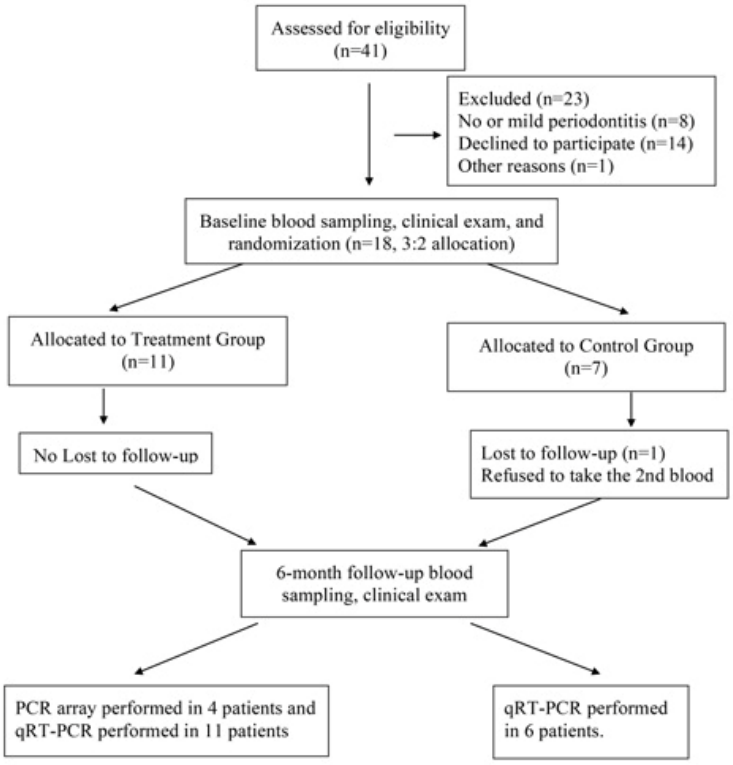
Table 3. Effects of periodontal treatment on the fold changes of IL-6 and IL-8 genes (log2 scale, mean ± SD) in EPCs from the Treatment (n=11) and Control groups (n=6)

Genes	Treatment group	Control group	Treatment effect (95% CI)
IL-6	-1.40 ± 1.27**	0.48 ± 0.96	-1.88 (-3.16 to -0.61) §§
IL-8	-0.78 ± 0.98 *	0.73 ± 1.49	-1.51(-2.78 to -0.24) §

Paired *t*-test for comparing intra-group difference, *$p<0.05$; **$p<0.01$.

Two-tailed independent *t*-test for comparing the inter-group difference, §$p<0.05$; §§$p<0.01$.



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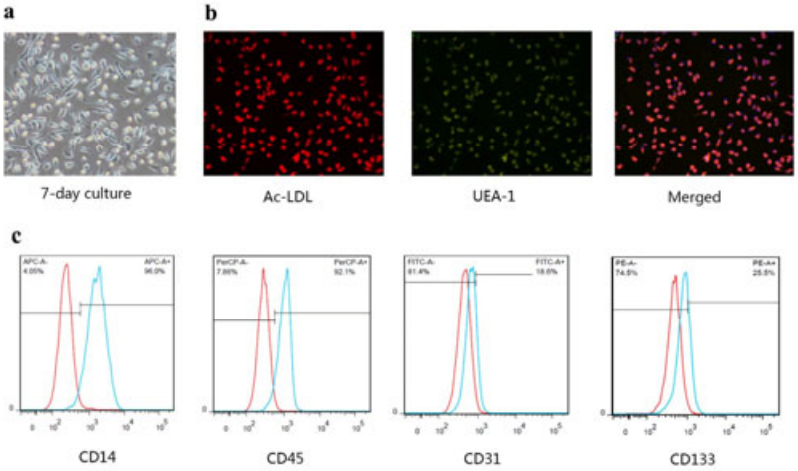

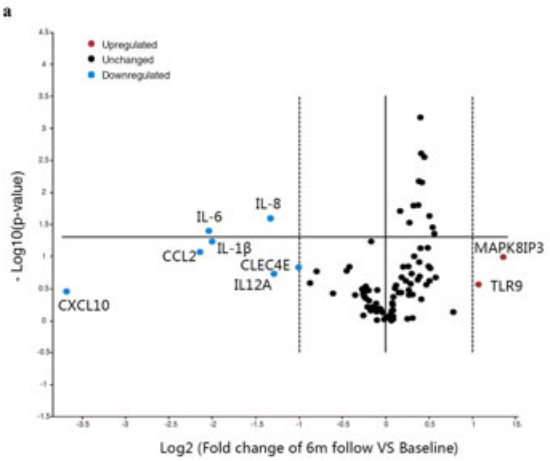

b

	GeneBank Accession No.	Description	Fold change	*p*-value
MAPK8IP3	NM_015133	Mitogen-activated protein kinase 8 interacting protein 3	2.56	0.103
TLR9	NM_017442	Toll-like receptor 9	2.10	0.273
CXCL10	NM_001565	Chemokine (C-X-C motif) ligand 10	-12.87	0.351
CCL2	NM_002982	Chemokine (C-C motif) ligand 2	-4.43	0.086
IL6	NM_000600	Interleukin 6 (interferon, beta 2)	-4.12	**0.040**
IL1b	NM_000576	Interleukin 1, beta	-4.02	0.059
CXCL8 (IL8)	NM_000584	Interleukin 8	-2.52	**0.026**
IL12A	NM_000882	Interleukin 12A	-2.45	0.187
CLEC4E	NM_014358	C-type lectin domain family 4, member E	-2.01	0.149

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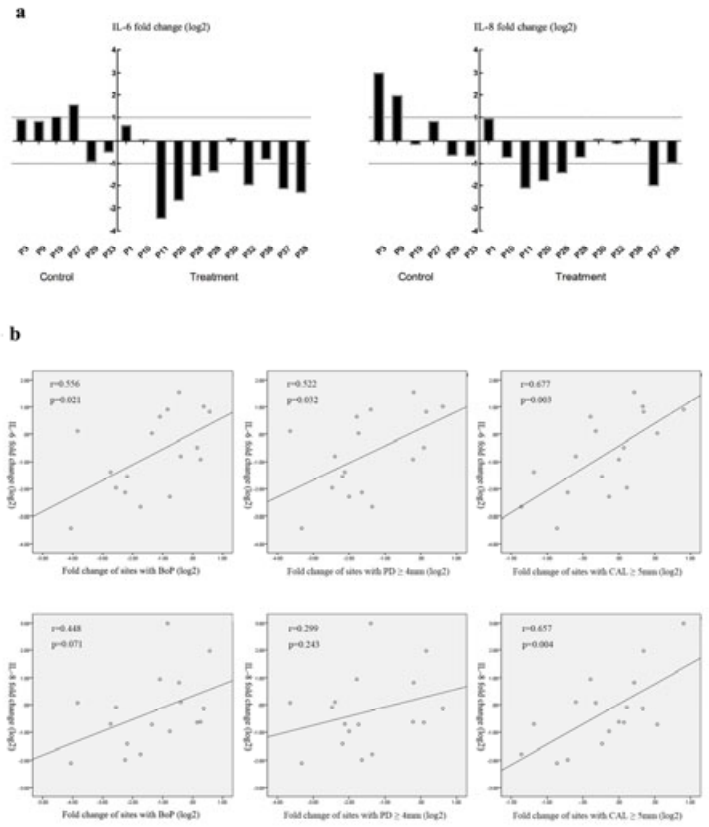

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